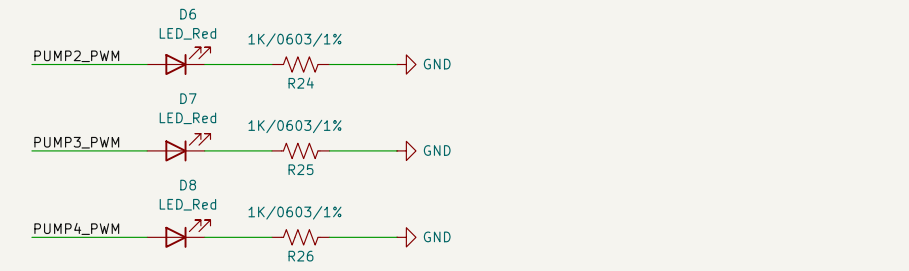
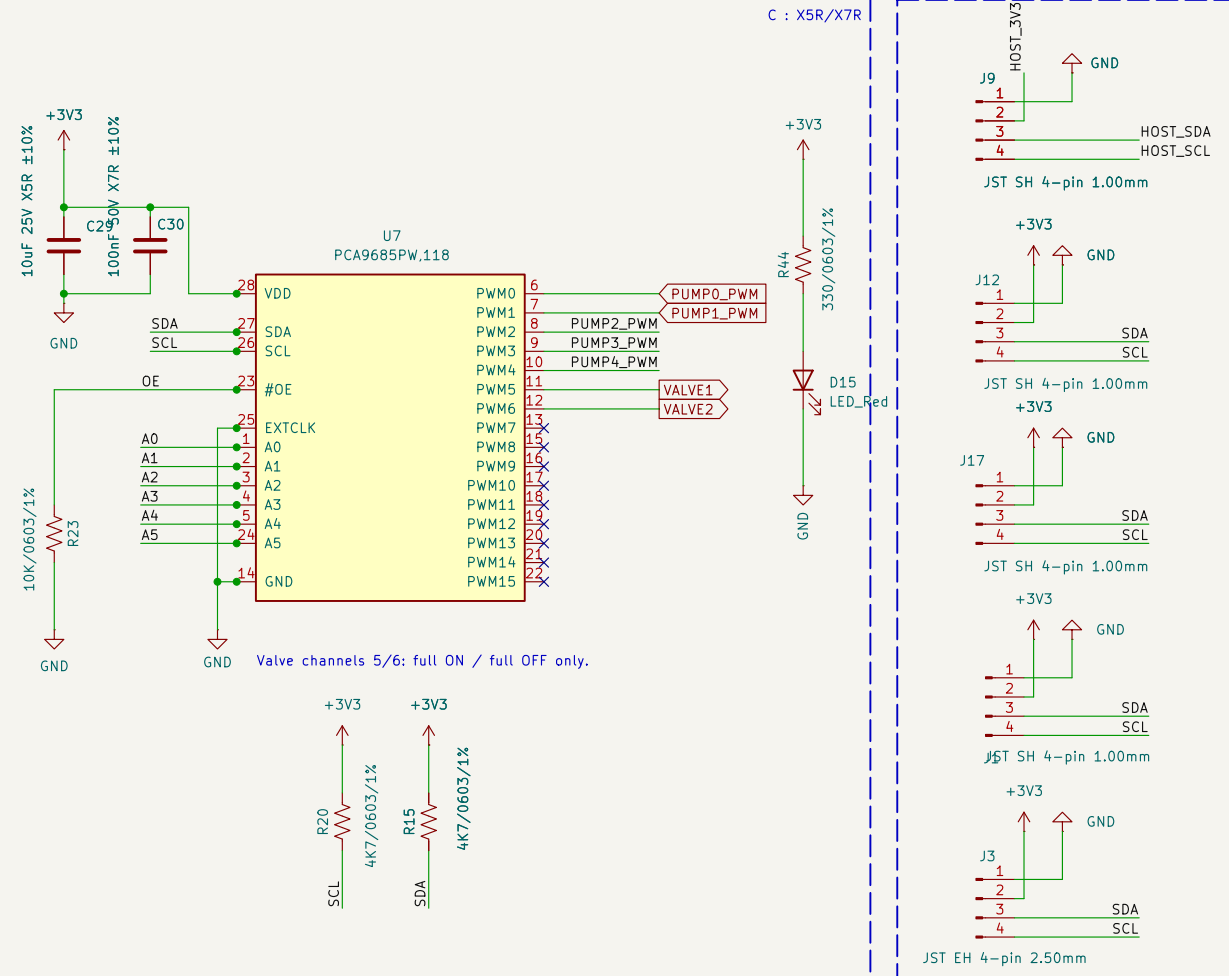
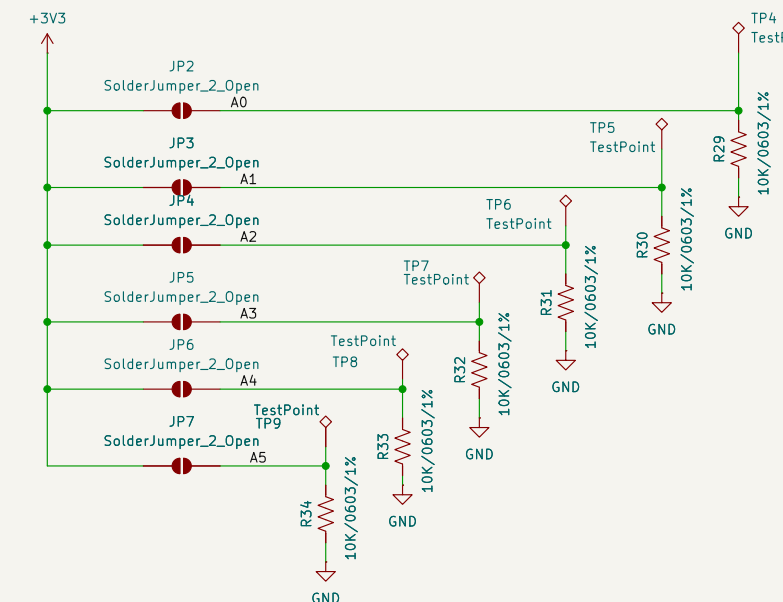
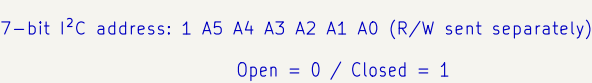
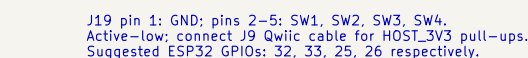
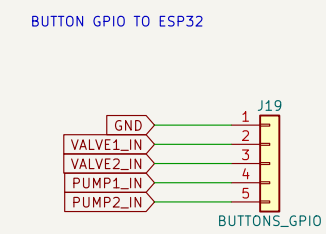
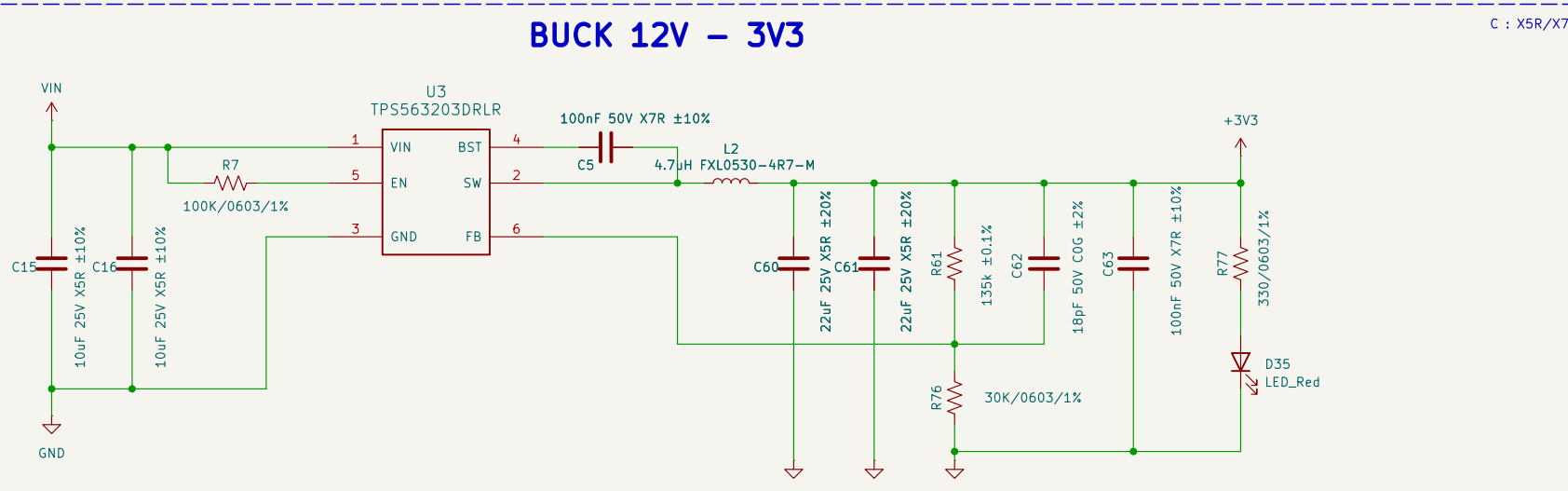
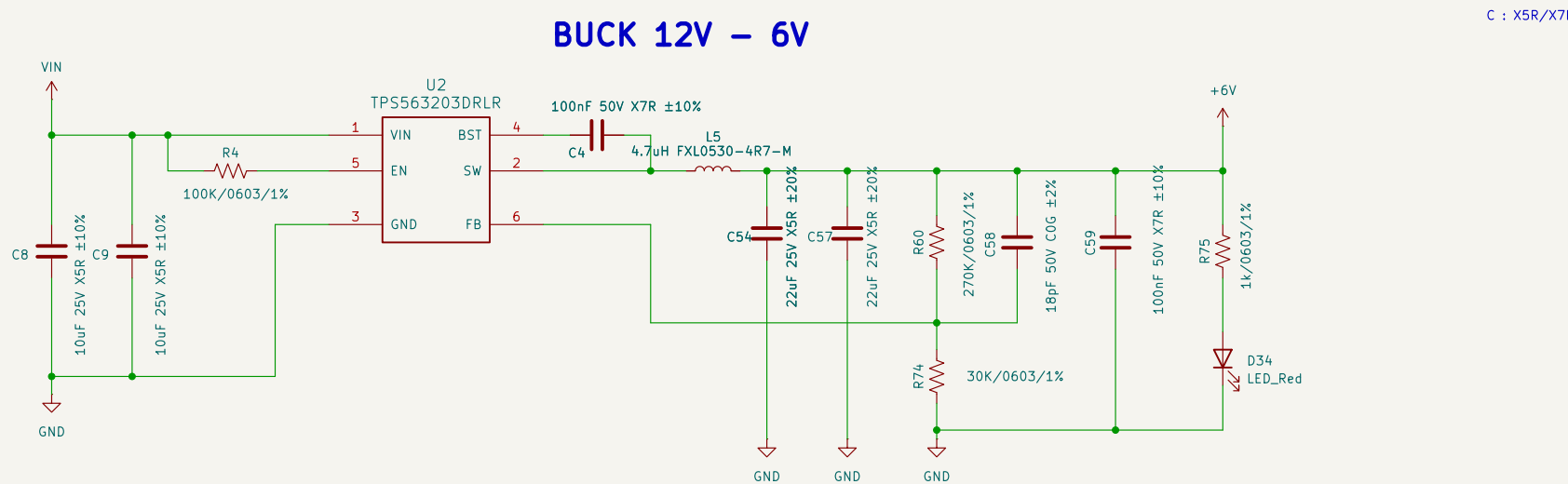
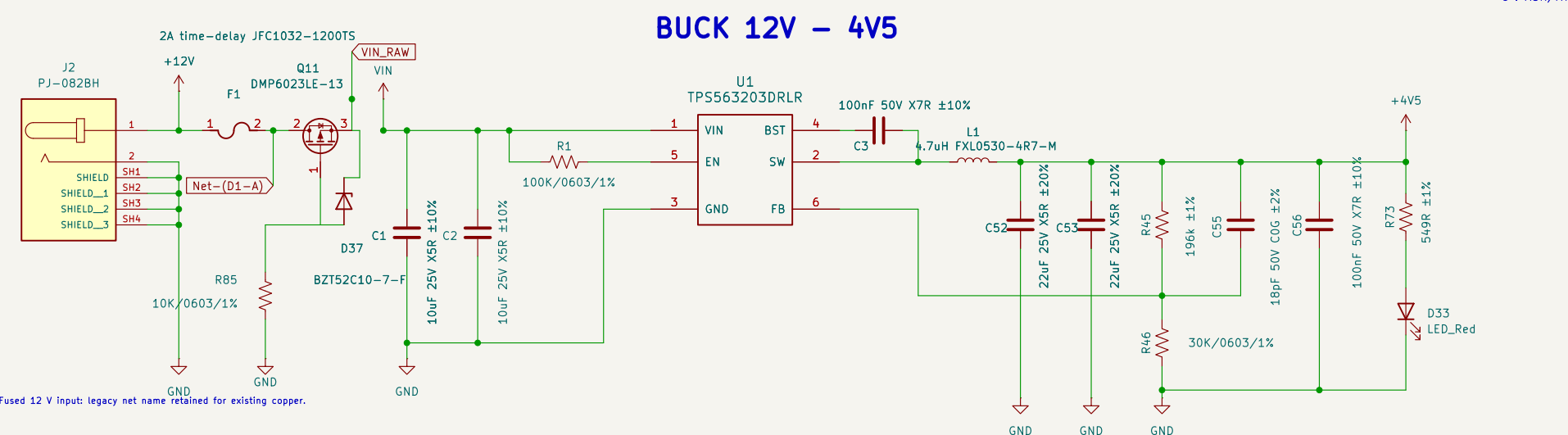
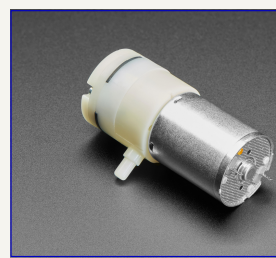
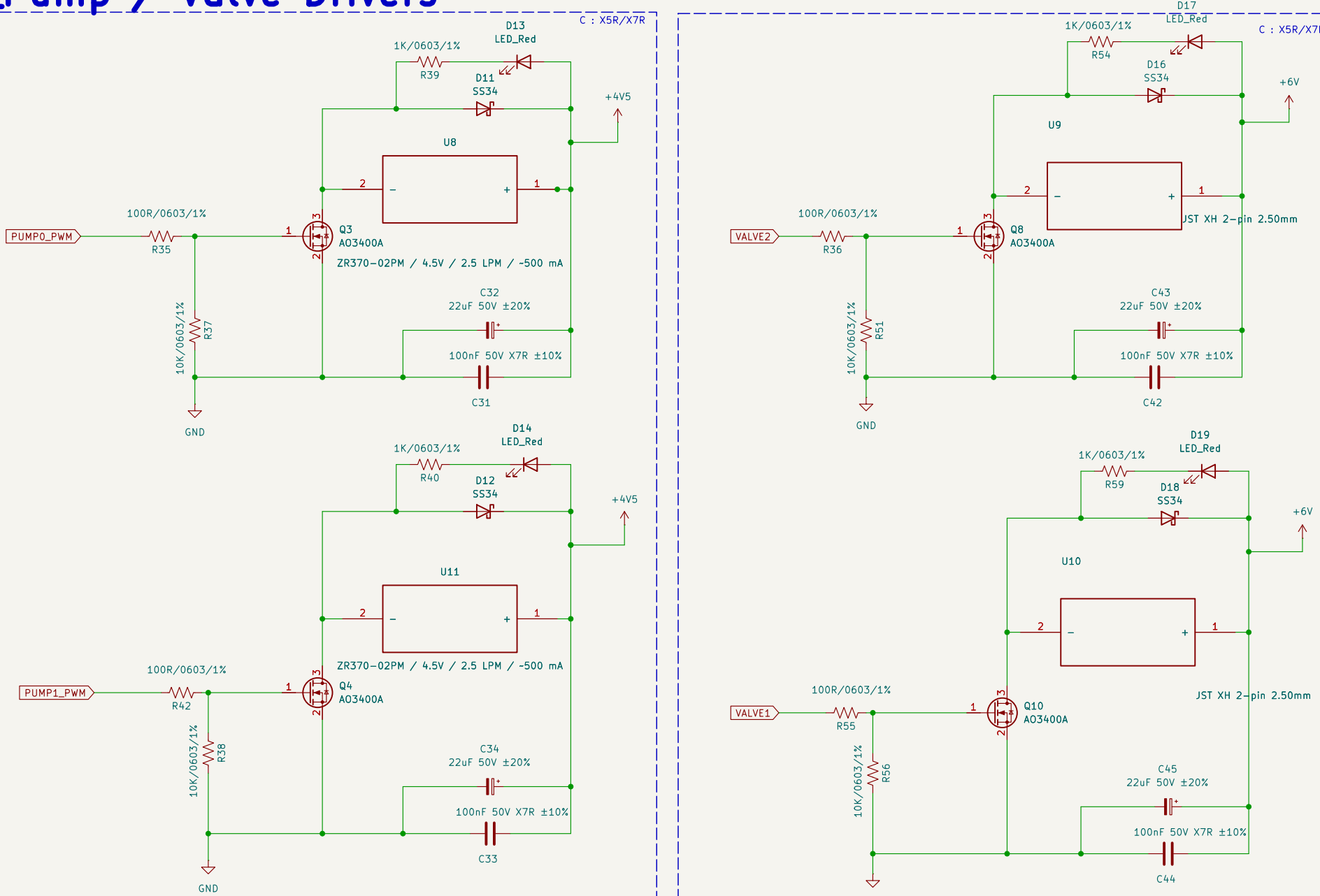


1_Power

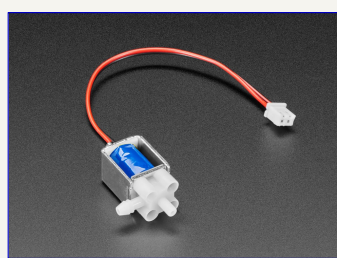


J3: LOCAL DEBUG BUS (GND / +3V3 / SDA / SCL): onboard supply.
J9: dedicated standard Qwiic master port. J1 / J12 / J17: powered sensors.
Multiple boards: external TCA9548A at 0x71: one channel active.
Disable PCA9685 ALLCALL after initialization.

2_Pump / Valve Drivers

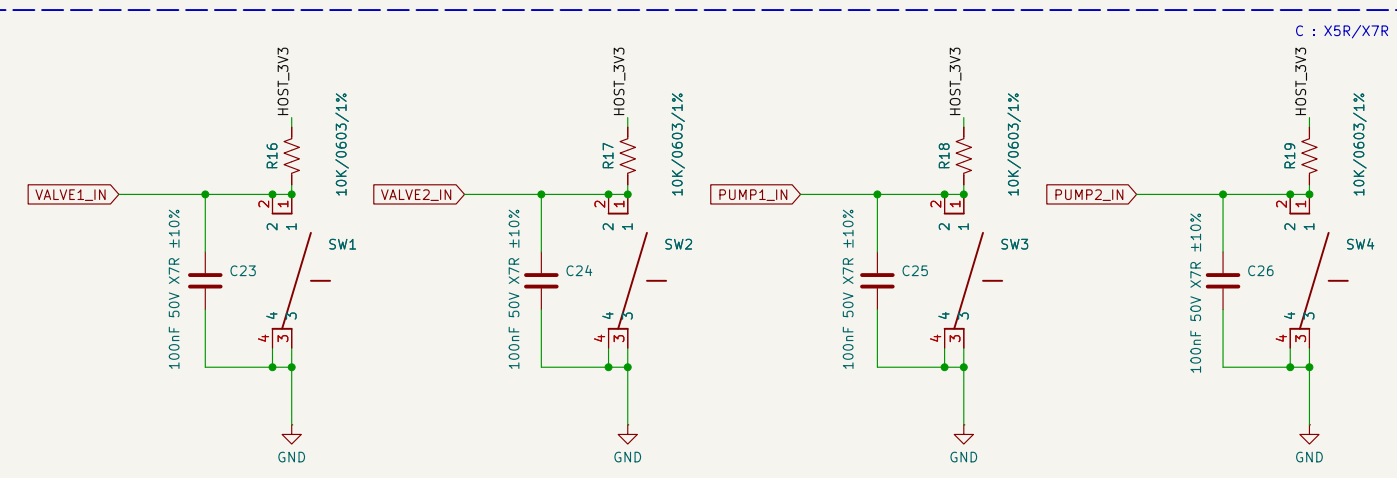


Rated voltage	DC:4.5V
Use voltage	DC:3.0V~5.0V
Load current	< 500mA

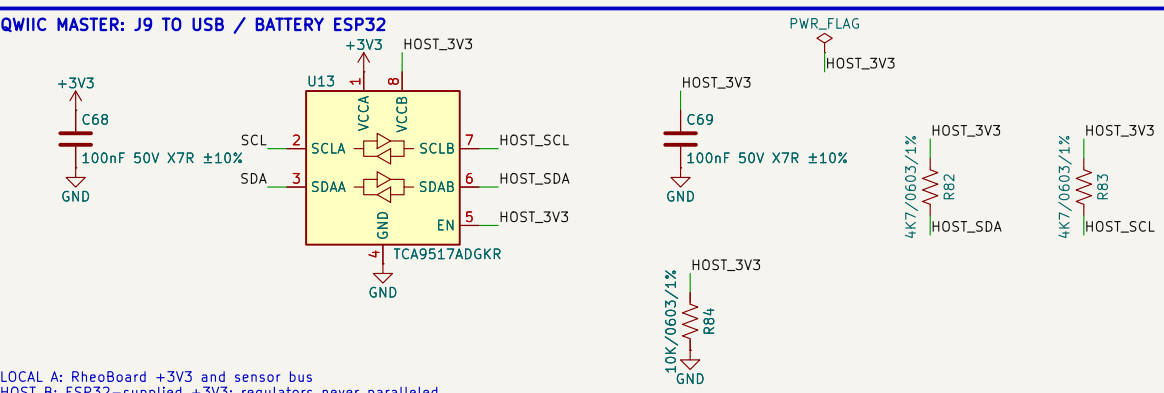
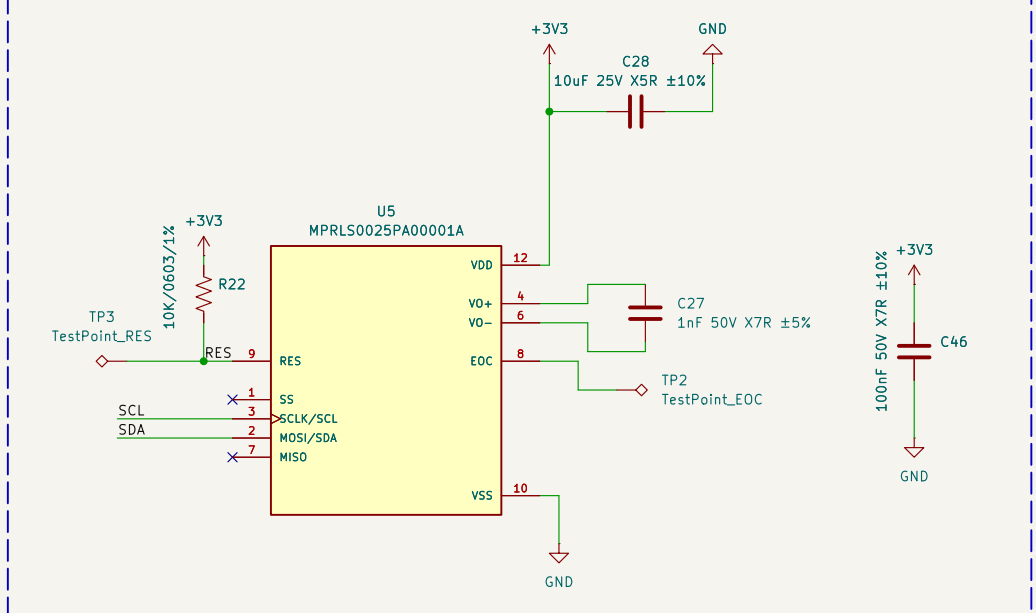


额定電壓 (Rated voltage) : DC 6.0V
使用電壓 (Operating voltage) : DC 5.0V~6.0V

4_Switch Inputs



5_Pressure Sensor



LOCAL A: RhoBoard +3V3 and sensor bus
HOST B: ESP32-supplied +3V3; regulators never paralleled
100 kHz initial bring-up; cable/module pull-ups share a current budget
J1 / J12 / J17 remain powered QWic sensor ports
J3 = LOCAL DEBUG BUS; no master-power isolation at that header

